Shun Ching Hsieh

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PROFILE

M.Ed. student in Educational Technology working at the intersection of **LLM-based educational systems, collaborative learning, and human-centered AI**. My research combines NLP technical work (transformer fine-tuning, RAG architecture, prompt engineering) with learning sciences theory (Knowledge Building, Socially Shared Regulation of Learning, OECD Learning Compass 2030) and practicum experience in K–12 classrooms. I am preparing for K–12 computer technology teaching in Taiwan and pursuing research on how AI can support, rather than displace, human regulatory and epistemic processes in learning.

EDUCATION

National Chengchi University (NCCU), Taiwan *2024 – 2026*
Master of Education, Educational Technology

- **Master's Thesis (Advisor: Prof. Huang-Yao Hong):** "Knowledge Building and OECD Learning Compass–Aligned Analysis of Competency-Based Lesson Plans"
- **Selected coursework:** Data Mining (National Tsing Hua University), Computational Thinking and AI, Educational Research Methodology, Mixed Methods Research, AI Ethics, Knowledge Creation and Educational Technology, Educational Technology Design, Large-Scale Educational Assessment Analytics

National Chengchi University (NCCU), Taiwan *2020 – 2024*
Bachelor of Education

RESEARCH EXPERIENCE

Research Assistant — Knowledge Building Lab *Sep 2024 – Jun 2026*
National Chengchi University, Taiwan

- Designed and built a Retrieval-Augmented Generation (RAG) workflow for analyzing and revising K–12 lesson plans aligned with Taiwan's 108 Curriculum Guidelines.
- Fine-tuned a RoBERTa-based classifier using HuggingFace Transformers to categorize instructional objectives across cognitive, affective, and psychomotor domains; achieved $\kappa = 0.83$ inter-rater reliability.
- Designed annotation criteria integrating Bloom's Taxonomy, CASEL Social-Emotional Learning competencies, and psychomotor learning frameworks.
- Trained and evaluated models on 800 manually annotated instructional objectives using human-in-the-loop validation workflows.
- Processed and analyzed 50,000+ educational records using Python automation pipelines (pandas, scikit-learn, transformer-based NLP).
- Investigated explainable AI approaches that preserve teacher agency through editable curriculum revision workflows.

Master's Thesis Research *2025 – 2026*
Advisor: Prof. Huang-Yao Hong | National Chengchi University

- Compared Taiwan's 108 Curriculum lesson plans with Knowledge Building lesson plans ($n = 133$).
- Operationalized OECD Learning Compass 2030 across three dimensions: Agency (AG), Anticipation–Action–Reflection (AAR), and Transformative Competencies (TC).
- Built a rubric-based analytical framework for evaluating collaborative and competency-oriented instructional design.
- Found that AAR-cycle quality showed the largest effect size in differentiating high-quality collaborative lesson plans — confirming epistemic cycle quality, not content coverage, distinguishes effective collaborative design.
- Investigated how reflection structure, epistemic cycles, and collaborative regulation shape instructional design outcomes.

Junior High School Teaching Practicum (Computer Technology) *Jan 2025 – Present*
Teacher-in-training under Taiwan's 108 Curriculum

- Observed how collaborative learning succeeds and breaks down in authentic classroom environments.
- Examined real-time student behaviors including disengagement, conversational derailment, surface compliance, and collaborative problem-solving — the same behavioral conditions targeted by recent AI-tutor research.
- Develops domain expertise in classroom group dynamics directly relevant to multi-party AI agent design and learner-persona research.

PUBLICATIONS & PRESENTATIONS

AERA 2026 Annual Meeting — Accepted Oral Presentation *Apr 2026*

American Educational Research Association | Los Angeles, California, USA

Hsieh, S. C. (2026, April). *Integrating RAG-Enhanced AI to Support Cognitive, Affective, and Psychomotor Balance in K–12 Lesson Plans* [Oral presentation]. AERA Annual Meeting, Los Angeles, CA.

- Developed a RAG-enhanced AI system using a fine-tuned RoBERTa classifier (HuggingFace Transformers) for K–12 curriculum analysis and lesson-plan revision.
- Achieved inter-rater reliability of $\kappa = 0.83$ between human coders on 800 annotated teaching objectives.
- Analyzed 58 K–12 lesson plans using transformer-based NLP workflows.
- Increased Social-Emotional Learning (SEL) objective coverage from 17% to 32% after AI-supported augmentation ($t = 7.82$, $p < .01$).
- Funding: National Science and Technology Council (NSTC), Taiwan.

ADDITIONAL TECHNICAL PROJECTS

Brain-to-Text EEG Neural Decoding *Jun 2025*

Kaggle Project

- Processed EEG neural-signal datasets for text-decoding tasks; implemented baseline ML and deep-learning models for neural-signal classification.
- Conducted feature engineering, exploratory data analysis, and model evaluation.
- Tech stack: Python, scikit-learn, PyTorch, pandas.

Bike-Sharing Demand Prediction *May 2025*

Data Science Coursework

- Built time-series forecasting models predicting bike availability across campus stations.
- Engineered lag-based and rolling-window statistical features; improved RMSE and MAE relative to baseline forecasting approaches.
- Tech stack: Python, pandas, scikit-learn, matplotlib.

TECHNICAL SKILLS

Programming & Data Science: Python (proficient), R, SQL; pandas, NumPy, scikit-learn, Jupyter.

AI / NLP / LLMs: HuggingFace Transformers, RoBERTa fine-tuning, RAG pipeline design, prompt engineering, OpenAI API, Anthropic API, human-in-the-loop annotation workflows; PyTorch (coursework level); actively developing vLLM, TRL, and model-training proficiency.

Research Methods: Quantitative research, mixed-methods research, statistical modeling, social network analytics, qualitative coding, learning analytics.

Educational Theory: Knowledge Building (Scardamalia & Bereiter), Socially Shared Regulation of Learning (SSRL), OECD Learning Compass 2030, Bloom's Taxonomy, CASEL SEL framework, Taiwan 108 Curriculum competency framework.

Languages: Mandarin Chinese (native), English (professional working proficiency).